

EXPERIMENT 6

Apply K-Means and Hierarchical Clustering on Real Dataset

Introduction to Clustering

Clustering is an unsupervised machine learning technique used to group similar data points together based on their characteristics. Unlike supervised learning, clustering does not require labeled data. The goal is to discover hidden patterns or natural groupings within the dataset.

Clustering is widely used in:

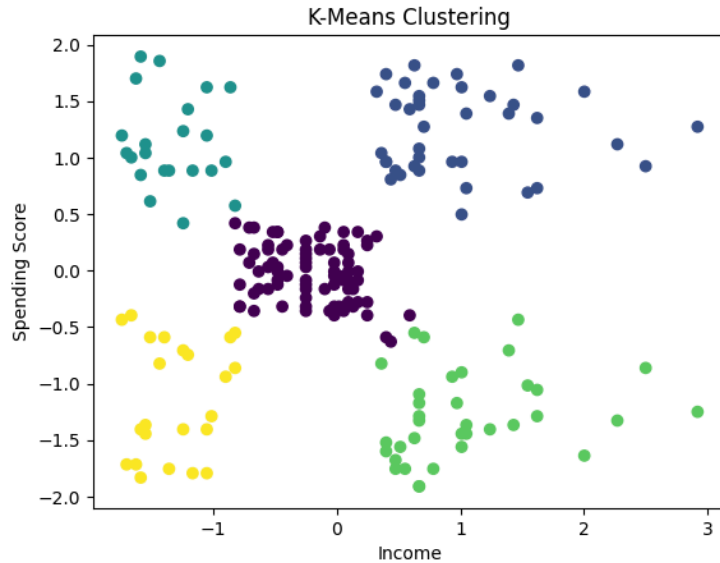
- Customer segmentation
- Market analysis
- Image segmentation
- Fraud detection
- Pattern recognition

K-MEANS CLUSTERING

K-Means clustering is one of the most popular unsupervised machine learning algorithms. Its goal is to group similar data points together and discover underlying patterns without prior labels.

What is K-Means?

K-Means is a partition-based clustering algorithm that divides the dataset into **K predefined clusters**. Each cluster is represented by its centroid (mean of the data points in that cluster). The algorithm minimizes the distance between data points and their respective cluster centroids.



Working of K-Means Algorithm

1. Select number of clusters (K)
2. Randomly initialize K centroids
3. Assign each data point to the nearest centroid
4. Recalculate centroids
5. Repeat until centroids stop changing

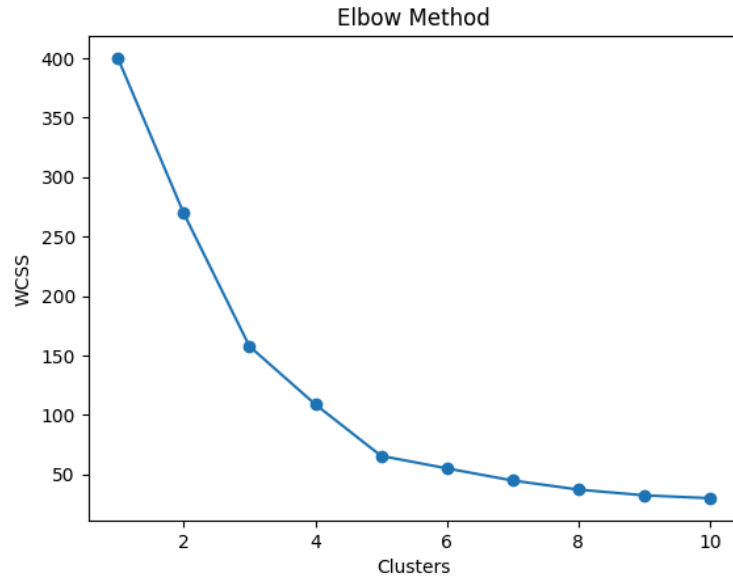
Elbow Method

The Elbow Method is used to determine the optimal value of K.

It plots:

- Number of clusters (K)
- WCSS (Within Cluster Sum of Squares)

The point where the graph forms an “elbow” indicates the optimal number of clusters.



Advantages of K-Means

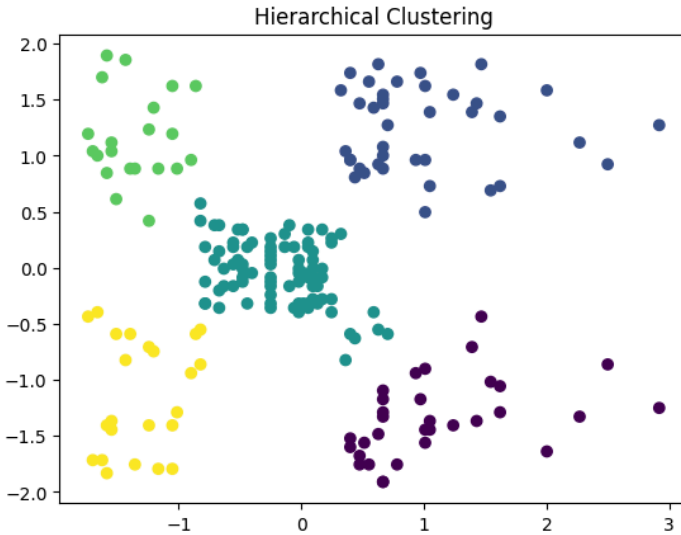
- Simple and easy to implement
 - Efficient for large datasets
 - Fast convergence
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Limitations of K-Means

- Requires predefined value of K
 - Sensitive to initial centroid selection
 - Works poorly for non-spherical clusters
 - Sensitive to outliers
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HIERARCHICAL CLUSTERING

Hierarchical Clustering builds a hierarchy of clusters using a tree-like structure called a dendrogram. It does not require specifying the number of clusters initially.



Types of Hierarchical Clustering

- Agglomerative (Bottom-up approach)
- Divisive (Top-down approach)

In this experiment, **Agglomerative Clustering** is used.

Working of Agglomerative Clustering

1. Start with each data point as its own cluster
 2. Find the two closest clusters
 3. Merge them
 4. Repeat until all points form a single cluster
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Distance Measures Used

Common linkage methods:

- Single linkage
- Complete linkage
- Average linkage

- Ward's method (used in this experiment)

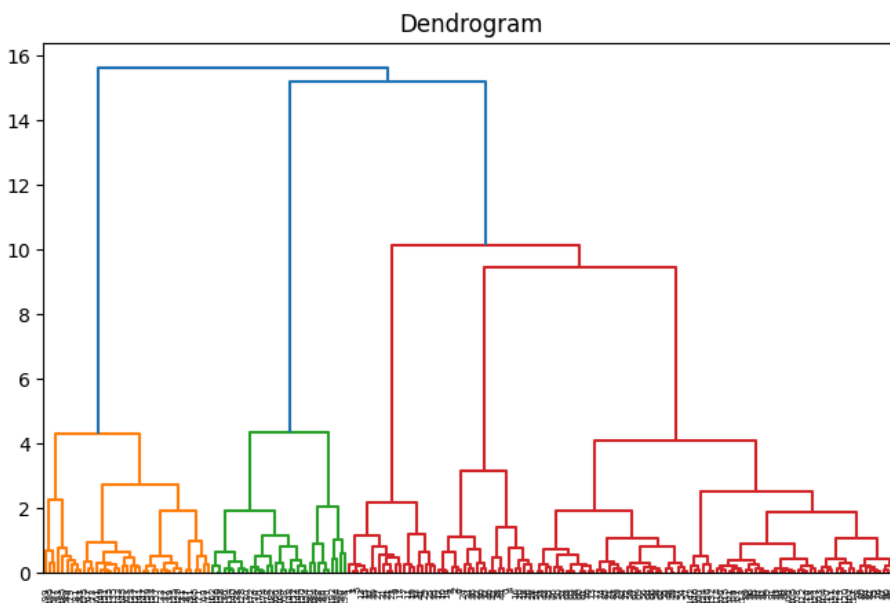
Ward's method minimizes variance and produces compact clusters.

Dendrogram

A dendrogram is a tree diagram that shows:

- How clusters are merged
- The distance at which clusters are combined

A horizontal cut across the dendrogram determines the number of clusters.



Advantages of Hierarchical Clustering

- No need to specify number of clusters initially
 - Produces interpretable tree structure
 - Useful for small datasets
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Limitations of Hierarchical Clustering

- Computationally expensive for large datasets
 - Once clusters are merged, they cannot be split
 - Sensitive to noise and outliers
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Comparison: K-Means vs Hierarchical Clustering

Feature	K-Means	Hierarchical
Type	Partition-based	Tree-based
Need to specify K	Yes	No
Speed	Faster	Slower
Scalability	Good for large data	Not ideal for large data
Visualization	Centroid-based	Dendrogram

1. Dataset Source

For this experiment, we used a **real-world dataset**:

- Dataset: *Mall Customer Segmentation Dataset*
- Source: Kaggle

Reason:

Unlike the synthetic dataset used in the original experiment (page 4), this dataset provides real customer behavior, making clustering more meaningful and practical.

2. Dataset Description

The dataset contains customer information collected from a shopping mall.

Features in Dataset

- Customer ID
- Gender
- Age
- Annual Income (k\$)
- Spending Score (1–100)

Features Used for Clustering

- Annual Income (k\$)
- Spending Score (1–100)

Dataset Characteristics

- Number of samples: 200
- Number of features used: 2
- No target variable (unsupervised learning)

Each data point represents a customer with:

- Income level
 - Spending behavior
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3. Performance Analysis

Since clustering is an unsupervised learning technique, accuracy cannot be used. Instead, we evaluate performance using:

K-Means Evaluation

- **Inertia (WCSS):** Measures compactness of clusters (lower is better)

- **Elbow Method:** Used to find optimal K (K = 5 observed)
 - **Silhouette Score:** Measures cluster separation (closer to 1 is better)
 - **Visualization:** Scatter plot shows clearly defined customer segments
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Hierarchical Clustering Evaluation

- **Dendrogram:** Used to determine number of clusters
 - **Silhouette Score:** Indicates quality of clustering
 - **Visualization:** Shows hierarchical grouping of customers
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Observation

Both algorithms successfully formed meaningful clusters such as:

- High income – high spending customers
- Low income – low spending customers
- Moderate income groups

This helps in **customer segmentation and targeted marketing**.

4. Hyperparameter Tuning

Hyperparameter tuning improves clustering quality.

K-Means Tuning

- **n_clusters:** Tested using Elbow Method → Optimal K = 5
- **init:** Used **k-means++** for better centroid initialization
- **n_init:** Increased for stability

Impact:

- Reduced WCSS
 - Improved cluster separation
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Hierarchical Clustering Tuning

- `n_clusters`: Selected using dendrogram
- `linkage`: Used **Ward method**

Impact:

- Compact clusters
 - Better interpretability
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Final Conclusion

- K-Means is faster and suitable for large datasets
- Hierarchical clustering provides better visualization through dendrograms
- Both methods effectively segmented customers based on income and spending behavior
- Real-world datasets provide more practical insights compared to synthetic data